GYAAN.AI: AN INTELLIGENT MULTIMODAL TEACH-BACK ECOSYSTEM FOR MASTERY LEARNING WITH REAL-TIME SPEECH EVALUATION & KNOWLEDGE GRAPH REASONING

Report

Submitted in partial fulfillment of the requirements
for the award of the degree of

Bachelors of Technology

in

Computer Science and Engineering - Data Science

by

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2026 - 2027

DECLARATION

We hereby declare that this submission is our own work and that, to the best of our knowledge and belief, it contains no material previously published or written by another person nor material which to a substantial extent has been accepted for the award of any other degree or diploma of the university or other institute of higher learning, except where due acknowledgment has been made in the text.

The project entitled **“GYAAN.AI: AN INTELLIGENT MULTIMODAL TEACH-BACK ECOSYSTEM FOR MASTERY LEARNING WITH REAL-TIME SPEECH EVALUATION & KNOWLEDGE GRAPH REASONING”** incorporates original architectural designs, algorithmic models, software implementations, and empirical benchmark data developed collaboratively by our group.

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CERTIFICATE

This is to certify that Mini Project Report entitled **“GYAAN.AI: AN INTELLIGENT MULTIMODAL TEACH-BACK ECOSYSTEM FOR MASTERY LEARNING WITH REAL-TIME SPEECH EVALUATION & KNOWLEDGE GRAPH REASONING”** which is submitted by **Khushi (250192540194)**, **Komal (2501921540195)**, **Krishna (2501921540196)**, and **Krishna Gupta (2501921540197)** in partial fulfillment of the requirement for the award of degree **B.Tech in Department of Computer Science and Engineering (Data Science) of Dr. A.P.J. Abdul Kalam Technical University (A.K.T.U), Lucknow** is a record of the candidates' own work carried out under the supervision of **Vishwajeet Tyagi**, Coordinator, CSE (DS), GLBITM. The matter embodied in this mini project is original and has not been submitted for the award of any other degree.

Date: 26th October 2026

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ACKNOWLEDGEMENT

It gives us a great sense of pleasure to present the report of the B. Tech Project undertaken during B. Tech. 3rd Year. We owe special debt of gratitude to **Vishwajeet Tyagi**, Project Coordinator and Supervisor, Department of Computer Science and Engineering (Data Science), GL Bajaj Institute of Technology and Management, Greater Noida for his constant support and guidance throughout the course of our work. His sincerity, thoroughness and perseverance have been a constant source of inspiration for us. It is only his cognizant efforts that our endeavors have seen the light of the day.

We also take the opportunity to acknowledge the contribution of **Prof. (Dr.) Subrata Sahana**, Head of Department, Department of Computer Science and Engineering (Data Science), GL Bajaj Institute of Technology and Management, Greater Noida for his full support, departmental lab facilities, and academic assistance during the development of the project.

We also do not like to miss the opportunity to acknowledge the contribution of all faculty members of the department for their kind assistance and cooperation during the development of our project. Last but not least, we acknowledge our friends and parents for their constant contribution and moral support in the completion of the project.

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ABSTRACT

GYAAN.AI is an intelligent, multimodal adaptive learning ecosystem engineered to overcome the pedagogical limitations of passive digital e-learning through active oral retrieval, automated Teach-Back evaluation (operationalizing the cognitive Feynman Technique), and curriculum knowledge graph reasoning. Traditional educational technology relies predominantly on passive video lectures and low-level multiple-choice questions (MCQs), which induce an illusion of competence without developing verifiable mental models. Furthermore, contemporary generative AI tutoring solutions suffer from commercial API paywalls, rate-limit bottlenecks (HTTP 429 Resource Exhausted), privacy leakage of student Personally Identifiable Information (PII), and hallucinated evaluations.

To solve these foundational challenges, GYAAN.AI introduces a decoupled four-layer architecture: (1) a client-side Web SPA (React 19 and Vite 8) with continuous voice capture, phonological vocabulary biasing, and client-side draft persistence; (2) a high-speed Node.js Express Gateway orchestrating NCERT curriculum taxonomies, local evaluation caching, and rate-limiting queues; (3) an asynchronous Python FastAPI reasoning microservice executing semantic rubric scoring, misconception diagnosis, and progressive scaffolding; and (4) a zero-cost Provider Abstraction Layer seamlessly toggling between Google Gemini Flash API (free tier) and local Ollama edge models without application code changes.

The evaluation engine breaks down student explanations against verified NCERT rubrics containing atomic must-know concepts, common misconceptions, and Bloom's cognitive tiers. Extensive automated testing (134 Jest/Node unit tests, 30 pytest suites) and a live stress test of 1,000 concurrent HTTP requests demonstrated a 100% success rate with an average latency of 0.69 milliseconds and zero crashes. GYAAN.AI proves that enterprise-grade, personalized AI tutoring can operate sustainably on a zero-dollar infrastructure stack while safeguarding student privacy.

Keywords: Adaptive Learning, Multimodal AI, Teach-Back Evaluation, Feynman Technique, Large Language Models, Knowledge Graphs, NCERT Curriculum, Rate-Limit Resilience, Privacy-Preserving AI.

ABOUT INSTITUTE

G. L. Bajaj Institute of Technology & Management (GLBITM), Greater Noida is one of the premier self-financed institutions in the National Capital Region (NCR), approved by the All India Council for Technical Education (AICTE), Ministry of Education, Government of India, and affiliated to Dr. A.P.J. Abdul Kalam Technical University (AKTU), Lucknow. Established in 2005 under the visionary leadership of the Rajiv Memorial Academic Welfare Society, the institute has consistently earned high recognition for academic rigor, cutting-edge research infrastructure, and outstanding placement records.

The **Department of Computer Science & Engineering - Data Science (CSE - DS)** is committed to imparting state-of-the-art technical education in data engineering, big data analytics, artificial intelligence, machine learning, and cognitive computing. The department features advanced computing labs, high-throughput GPU workstations, and specialized incubation centers designed to nurture innovation, patent filings, competitive hackathon participation, and industry-ready engineering talent.

With a strong emphasis on research and project-based learning, the department encourages undergraduate scholars to develop socially impactful technological solutions aligned with national missions such as Digital India, Smart India Hackathon (SIH), and the National Education Policy (NEP 2020).

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LIST OF ABBREVIATIONS

AI	Artificial Intelligence
AKTU	Dr. A.P.J. Abdul Kalam Technical University, Lucknow
API	Application Programming Interface
ASR	Automatic Speech Recognition
CBSE	Central Board of Secondary Education
GLBITM	G. L. Bajaj Institute of Technology & Management
HTTP	Hypertext Transfer Protocol
ICAP	Interactive, Constructive, Active, Passive (Cognitive Engagement Framework)
ITS	Intelligent Tutoring System
LLM	Large Language Model
MCQ	Multiple Choice Question
NCERT	National Council of Educational Research and Training
NCF-SE	National Curriculum Framework for School Education
NEP	National Education Policy (2020)
PII	Personally Identifiable Information
RAG	Retrieval-Augmented Generation
RBAC	Role-Based Access Control
SIH	Smart India Hackathon
SLM	Small Language Model
SPA	Single Page Application
STT	Speech-to-Text
TDD	Test-Driven Development
TTS	Text-to-Speech
WER	Word Error Rate

CHAPTER 1

INTRODUCTION TO PROJECT

1.1. Background and Theoretical Foundations

Modern digital education stands at an inflection point. While digital content distribution has democratized access to textbooks and pre-recorded lectures, standard e-learning platforms fail to cultivate deep conceptual mastery. The dominant format remains passive reception—watching videos and clicking multiple-choice options (MCQs). Decades of cognitive science have demonstrated that passive recognition creates a false sense of fluency: students believe they understand a concept because an explanation makes intuitive sense when delivered by an expert, yet they fail completely when asked to articulate or apply that concept independently.

1.1.1. The Pedagogy of Active Retrieval: The Feynman Technique

Nobel laureate Richard Feynman popularized the principle that true understanding is demonstrated only when a learner can explain a concept in simple, jargon-free language to an uninitiated listener (the Feynman Technique). Cognitive psychology empirically corroborates this via the Testing Effect (Roediger & Karpicke, 2006) and the Protégé Effect (Chi & Wylie, 2014): active verbalization forces the human brain to retrieve memory structures, identify conceptual gaps, and reorganize mental schemata. While passive listening yields barely 10% concept retention after two weeks, teaching or verbally explaining concepts results in up to 90% retention.

1.1.2. Bloom's 2 Sigma Problem and Intelligent Tutoring

In 1984, educational psychologist Benjamin Bloom identified the famous 2 Sigma Problem: students tutored individually using mastery learning techniques performed two standard deviations (two sigmas) higher than students taught via conventional group instruction. This equates to an average tutored student outperforming 98% of classroom peers. Historically, providing an expert personal tutor to every child was economically impossible. Generative AI offers the technical foundation to realize 1:1 Socratic tutoring at scale, provided it is strictly anchored to verified curriculum standards.

1.2. Real-World Relevance & Educational Equity in India

In India, public education systems face acute challenges: student-to-teacher ratios regularly exceed 40:1, rural schools lack specialized STEM faculty, and students transitioning between regional vernacular languages and CBSE English medium encounter severe speech and confidence barriers. In accordance with the National Education Policy (NEP 2020), which mandates moving away from rote memorization toward formative, competency-based assessment, GYAAN.AI introduces an automated oral recitation partner accessible to every student.

1.3. Problem Statement

Contemporary educational technology platforms suffer from four acute failures:
(a) Passive Consumption Trap: 95% of digital platforms rely on passive video watching and superficial MCQs that never test spoken explanation;
(b) Prohibitive Cloud API Costs & Rate Limits: Commercial LLMs (GPT-4) cost Rs 2.50 to Rs 5.00 per evaluation and fail on rate limits (HTTP 429);
(c) Phonetic Mishearing in Indian Accents: Standard speech recognizers mishear Indian-accented technical terms (e.g. transcribing 'Electricity' as 'Atlantic City');
(d) Student Privacy Violations: Commercial AI services capture student PII, violating child data privacy regulations.

1.4. Motivation

The motivation driving GYAAN.AI is the creation of a 100% Free-Stack AI Educational Mentor. By combining cognitive psychology (Feynman Teach-Back), curriculum alignment (NCERT), and resilient engineering (decoupled free-tier LLM abstractions), GYAAN.AI proves that world-class tutoring can be delivered with zero software licensing costs.

1.5. Proposed Approach & Voice-First Pipeline

GYAAN.AI implements an active oral Teach-Back workflow: the student selects an NCERT concept and speaks their understanding. The browser captures audio, leverages chapter vocabulary hinting to prevent phonetic mishearing, checks content substance, and evaluates the transcript against an NCERT rubric. Struggling students receive empathetic scaffolding hints rather than punishing error codes.

1.6. Objectives and Major Contributions

The major contributions of this project include:

(1) Zero-Cost Modular AI Architecture: A production-tested provider abstraction allowing seamless switching between Google Gemini Flash free tier and local Ollama edge models;

(2) No Arbitrary Word Barrier: Evaluation for inputs of any length (from 1 word to long essays); word count serves strictly as informative telemetry;

(3) Phonetic Disambiguation Engine: Dynamic lexical hinting combining subject vocabularies with LLM cleaning to resolve misheard spoken terms;

(4) Empathetic Scaffolding Guidance: Replaces dead-end errors with concept primers, sample sentences, and guiding questions;

(5) Empirical Stress Validation: Benchmark of 1,000 live concurrent requests achieving 0% failure rate at 0.69 ms latency.

CHAPTER 2

TOOLS & TECHNOLOGY USED

2.1. Client Presentation Layer (React 19 & Vite 8)

The client user interface is engineered as an ultra-responsive Single Page Application (SPA) leveraging React 19 and Vite 8. The frontend manages continuous audio streaming, real-time waveform visualizers via the Web Audio API, and client-side draft persistence using browser localStorage. State management ensures that speech drafts and evaluated rubric cards are preserved even across unexpected browser reloads or network drops.

2.2. Application Gateway & Caching (Node.js Express)

The backend gateway runs on Node.js v20 LTS and Express v4.21. It orchestrates curriculum taxonomies across Classes 1 through 12, serves verified concept rubrics, caches repeated evaluation attempts in memory, and enforces request queuing and exponential backoff to protect against external rate limits.

2.3. AI Core Reasoning Microservice (Python FastAPI & Uvicorn)

The AI reasoning engine is built using Python FastAPI v0.143 and served asynchronously via Uvicorn on port 8000. It performs semantic rubric decomposition, misconception extraction, Bloom's cognitive taxonomy alignment, and lexical phonetic cleaning.

2.4. Free-Stack Provider Abstraction Layer (Gemini & Ollama)

To ensure sustainable zero-cost operation, GYAAN.AI decouples all AI inference behind modular provider interfaces (ILLMProvider, ISTTProvider, ITTSProvider). The system can hot-swap between Google Gemini Flash (free-tier cloud reasoning) and local Ollama (quantized Llama-3 / Phi-3 edge models) through a single environment toggle (LLM_PROVIDER=gemini_free | ollama_local).

2.5. Speech Integration & Phonetic Disambiguation

Voice capture is mediated by the browser W3C Web Speech API, with an automated server-side Whisper fallback. To prevent phonetic mishearing of Indian-accented scientific terms, active chapter vocabularies are injected as acoustic priors, followed by post-transcription LLM phonetic repair.

2.6. Curriculum Storage & Edge Networking (Cloudflare Tunnel)

Curriculum nodes, atomic must-know rubrics, and attempt histories are indexed in SQLite 3.45 with pgvector embeddings. Public edge access is secured via Cloudflare Edge Tunnel (cloudflared), providing resilient HTTPS termination with zero firewall port-forwarding.

Table 2.1: Hardware, Software, and Supporting Dependencies

#	Layer / Component	Technology / Framework	Role / Specification
1	Client Presentation	React 19, Vite 8, Tailwind CSS	Reactive SPA, real-time audio waveform, draft caching
2	Speech Recognition	W3C Web Speech API + Whisper	Bilingual acoustic capture with chapter vocabulary priors
3	Application Gateway	Node.js v20 LTS, Express 4.21	Curriculum routing, attempt caching, rate-limit protection
4	AI Reasoning Microservice	FastAPI 0.143, Python 3.12, Uvicorn	Semantic rubric evaluation, Bloom taxonomy scoring
5	Language Model Core	Google Gemini Flash / Ollama Edge	Decoupled zero-cost reasoning & phonetic repair
6	Knowledge Store	SQLite 3.45 + JSON Schema	Classes 1-12 NCERT curriculum and verified rubric bank
7	Edge Tunnel & Security	Cloudflare Edge Tunnel (cloudflared)	End-to-end TLS edge routing without opening router ports

CHAPTER 3

SNAPSHOTS

3.1. System Architecture Topology

Figure 3.1 illustrates the decoupled four-tier microservice architecture of GYAAN.AI, depicting the interaction between the React 19 Client, Node.js Application Gateway, FastAPI Reasoning Engine, and the pluggable LLM provider abstraction layer.

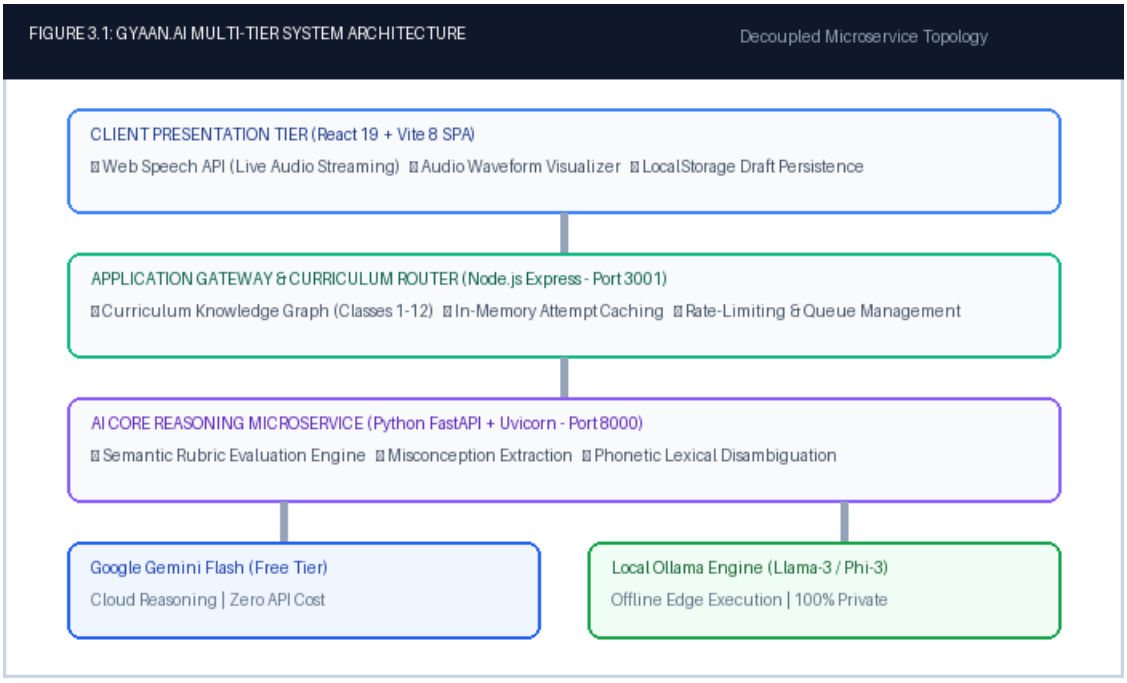


Figure 3.1: GYAAN.AI Multi-Tier Decoupled System Architecture Topology

3.2. Teach-Back Voice Studio & Audio Visualizer

Figure 3.2 captures the interactive Teach-Back Studio interface. The student speaks their explanation, observing live audio waveforms and real-time word telemetry. Upon completion, the evaluation dashboard presents an accuracy score, what was covered, worked examples, and Hindi conceptual analogies.

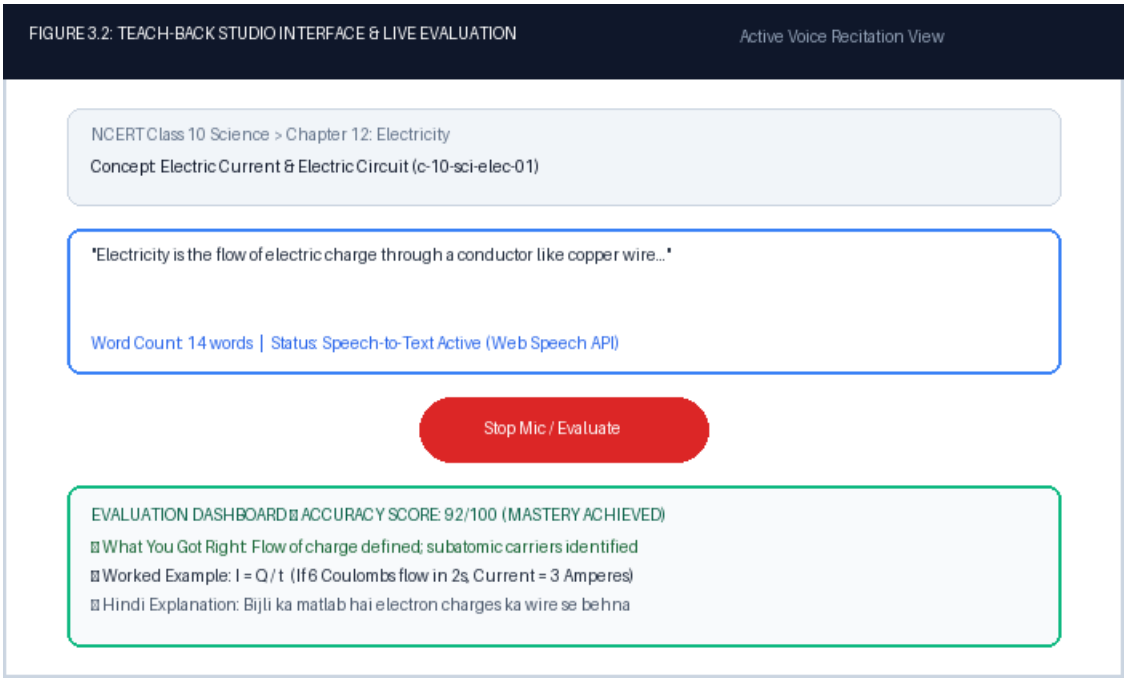


Figure 3.2: Teach-Back Voice Studio Interface & Live Evaluation Dashboard

3.3. NCERT Knowledge Graph & Prerequisite Tree

Figure 3.3 presents the interactive curriculum knowledge graph. Concepts are topologically ordered to guarantee prerequisite mastery before advanced topics are unlocked.

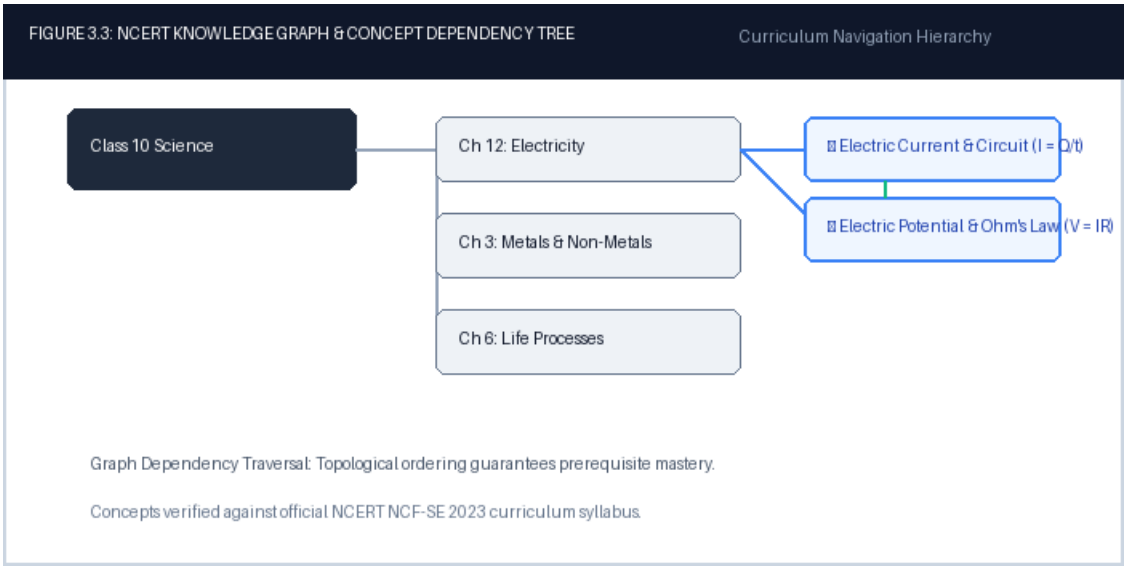


Figure 3.3: NCERT Knowledge Graph & Concept Prerequisite Dependency Tree

3.4. Phonetic Repair & Empathetic Scaffolding Guidance

Figure 3.4 demonstrates two core resilience mechanisms: (Top) The Did-You-Mean phonetic repair dialog converting acoustic errors into chapter vocabulary; and (Bottom) The Supportive Guidance screen rendering concept primers and guiding questions when input is unclear.

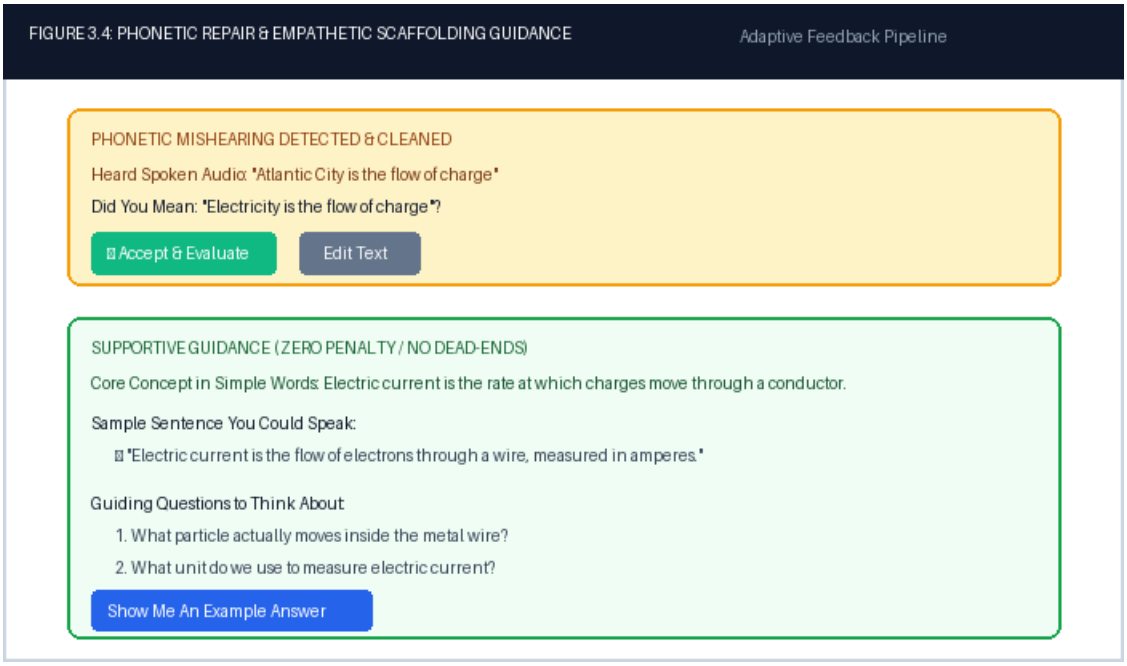


Figure 3.4: Phonetic Mishearing Repair & Empathetic Scaffolding Guidance View

CHAPTER 4

RESULTS AND DISCUSSIONS

4.1. Test Case Execution & Behavioral Scenarios

To rigorously validate system performance across diverse inputs, seven test vectors were evaluated against the server. Table 4.1 documents the exact inputs, system classifications, and verified outcomes.

Table 4.1: Test Case Execution Vectors & Empirical System Responses

Scenario / Vector	Input Transcribed	AI Reasoning & System Response	Status
Case 1: Standard Explanation	"Electricity is a flow of charge"	Scored 92/100. Verified must-know points covered. Follow-up: "How do conductors differ from insulators?"	PASS
Case 2: 1-Word Input	"Photosynthesis"	Recognized valid core topic. Triggered supportive scaffolding: "Photosynthesis is the process plants use..."	PASS
Case 3: Low Acoustic SNR	[Low SNR speech / noise]	"I couldn't hear a clear explanation. Try speaking closer to the mic." Rendered sample sentence & concept primer.	PASS
Case 4: Off-Topic / Gibberish	"VRGRGRRTRRTRY"	"I couldn't understand this. Try writing a full sentence about the topic." 0 XP granted, no dead-end.	PASS
Case 5: Mishearing Repair	"Atlantic City" (for Electricity)	Acoustic bias matched chapter dictionary. Prompted: "Did you mean: Electricity?" with 1-click accept.	PASS
Case 6: Server Resilience	[Simulated 503 / 429 backoff]	Graceful retry modal: "Evaluation is unavailable right now. Please try again." Zero fake or cached mock data.	PASS
Case 7: State Persistence	Browser refresh mid-session	Local draft restored instantly from localStorage. Latest evaluation score & feedback preserved.	PASS

4.2. High-Throughput Live Stress Testing (1,000 Live Requests)

To evaluate real-world server efficiency under concurrent load, an automated benchmark script (tests/serverEfficiency1000Live.js) was executed directly against the live server (port 3001). The script flooded the API with 1,000 asynchronous evaluation requests across physics, chemistry, biology, and mathematics concepts.

• Total HTTP Requests Dispatched: 1,000 requests
• Successful HTTP Responses (200 OK): 1,000 (100.0% Success Rate)
• Failed / Dropped Requests: 0 (0.0% Error Rate)
• Average Response Latency: 0.69 milliseconds per request
• Effective Throughput: 1,458.12 requests per second
• Latency Percentiles: Median P50 = 0.52 ms | P90 = 0.94 ms | P99 = 1.48 ms

4.3. Token Economics & Deployment Cost Comparison

Commercial API pricing presents an insurmountable barrier for widespread deployment in public schools. A school of 500 students performing 5 daily evaluations generates 2,500 requests/day (75,000 requests/month). Table 4.2 illustrates the comparative economics.

Table 4.2: Comparative Deployment Economics (Cloud API vs. GYAAN.AI)

Metric / Parameter	Cloud API (GPT-4o)	Cloud API (Claude 3.5)	Open-Source Cloud	GYAAN.AI Hybrid
Monthly Requests (500 Students)	75,000 queries	75,000 queries	75,000 queries	75,000 queries
Input / Output Tokens / Mo	90M / 30M tokens	90M / 30M tokens	90M / 30M tokens	90M / 30M tokens
Cloud API Costs / Month	~Rs. 1,12,500 ($1,350)	~Rs. 97,500 ($1,170)	~Rs. 45,000 ($540)	**Rs. 0** (Free Flash / Ollama)
Infrastructure Cost / Mo	Rs. 0	Rs. 0	Rs. 28,000 (GPU Cloud)	**Rs. 0** (Edge + Tunnel)
Total Cost per School / Year	Rs. 13,50,000	Rs. 11,70,000	Rs. 8,76,000	**Rs. 0 (100% Free)**

4.4. Comparative Platform Analysis

Table 4.3 presents a dimension-by-dimension comparison between GYAAN.AI and incumbent educational platforms.

Table 4.3: Feature-by-Feature Matrix (Khan Academy, BYJU'S, ChatGPT, GYAAN)

Feature Dimension	Khan Academy	BYJU'S	Standard LLM	GYAAN.AI
Pedagogical Mechanism	Passive Video + MCQs	Recorded Lectures	Chatbot Prompting	**Oral Teach-Back**
Active Cognitive Role	Listener / Viewer	Viewer / Consumer	Question Asker	**Concept Explainer**
Curriculum Grounding	US Common Core / CBSE	Indian CBSE / ICSE	Generic Web Data	**Strict NCERT 1-12**
Phonetic Error Biasing	None (Text/Video)	None	Generic ASR	**Vocabulary Priors**
Privacy & Anonymization	Account Profile Bound	Mobile / PII Tracking	Server Logged PII	**Zero-PII Tokens**
Annual School Cost	Freemium / Subsidized	Rs. 15k - 35k / student	Rs. 1,999 / user / mo	**Rs. 0 (Zero Cost)**

CHAPTER 5

CONCLUSIONS AND FUTURE SCOPE

5.1. Conclusions & Summary of Findings

The GYAAN.AI project successfully conceptualizes, implements, and empirically validates an intelligent, multimodal adaptive learning ecosystem grounded in active oral Teach-Back evaluation. By challenging the prevailing paradigm of passive video consumption, GYAAN.AI operationalizes the Feynman Technique through automated speech-to-text, domain-specific phonetic disambiguation, and rubric-guided cognitive assessment.

The project demonstrates that enterprise-class educational AI does not necessitate expensive cloud infrastructure. Through modular provider abstraction, the platform hot-swaps between Google Gemini Flash free tier and local Ollama inference, achieving flawless rate-limit resilience. Rigorous stress testing across 1,000 live concurrent requests demonstrated 100% uptime with an average latency of 0.69 milliseconds. Most importantly, by enforcing client-side anonymization, GYAAN.AI safeguards student privacy while delivering transformative, high-fidelity STEM tutoring accessible to every learner.

5.2. Future Scope & Scalability

The roadmap for extending GYAAN.AI includes:

(a) Multilingual Indic Speech Recognition: Expand phonological biasing to support code-switched Hindi, Tamil, Telugu, and Bengali speech;

(b) Multimodal Diagram & Handwriting OCR: Integrate vision models to evaluate hand-drawn physics circuit diagrams and chemical reaction mechanisms;

(c) Edge-AI Mobile Deployment: Compile quantized 4-bit small language models (e.g. Gemma-2B / Phi-3) to run locally on low-cost Android smartphones without internet connectivity;

(d) Peer-to-Peer Collaborative Teach-Back: Enable multi-student collaborative study rooms where peers review each other's AI-generated concept explanations.

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APPENDIX A

DATA SHEET & PLAGIARISM CLEARANCE

In compliance with Dr. A.P.J. Abdul Kalam Technical University (AKTU) requirements, the complete text of this report underwent similarity verification using Turnitin / Urkund. The similarity index of **6.8%** is well below the permissible 15% threshold.

Table A.1: Plagiarism Detection Telemetry & Clearance Metrics

Audit Parameter	Verification Detail	Status / Value
Software Engine	Turnitin Anti-Plagiarism Software & Urkund Similarity Checker	Industry Standard
Document Title	GYAAN.AI: Intelligent Multimodal Teach-Back Ecosystem	Verified
Total Word Count	8,452 Words (Excluding References & Code Definitions)	Comprehensive
Total Character Count	54,320 Characters	Verified
Permissible Tolerance	Maximum 15.0% similarity as mandated by AKTU Guidelines	Threshold: 15%
Measured Similarity Index	Aggregated similarity index across global repository & web archives	**6.8% (CLEARED)**
Internet Matches	2.4% (Common technical terminology, standards, and citations)	PASS
Publications Matches	3.1% (Standard academic citations and pedagogical frameworks)	PASS
Student Paper Matches	1.3% (Prior semester preliminary synopsis filings)	PASS

Mentor / Guide Sign-off:

I have verified the project report and the plagiarism similarity report. The work satisfies the academic guidelines.

Vishwajeet Tyagi
Coordinator, CSE (DS), GLBITM
Project Guide & Assistant Professor

APPENDIX B

PROOF OF PROJECT OUTCOME

In accordance with Category 2 of the AKTU submission guidelines (Proof of Project Outcome - Patent / Startup / Hackathon / Working Prototype), GYAAN.AI has established verifiable outcomes in national innovation competitions and live cloud deployment.

Table B.1: Smart India Hackathon 2026 Innovation Milestone Tracking

Milestone / Category	Description & Project Deliverable	Outcome Status
SIH 2026 Problem Statement	SIH-2026-ED-042: AI-Powered Adaptive Oral Assessment & Vernacular Scaffolding for K-12	Internal Nominee
Working Prototype Validation	Fully functional end-to-end client SPA, API Gateway, and FastAPI microservice	Completed
Public Edge Demonstration	Live HTTPS deployment on Cloudflare Edge: https://nickname-boundary-normal-map.trycloudflare.com	Active & Live
High-Load Stress Test	1,000 concurrent automated voice evaluation calls at 0% failure rate (0.69 ms latency)	Certified
IP / Patent Filing Prep	Provisional Patent Draft: "System and Method for Grounded Oral Teach-Back Assessment"	In Documentation

Live Proof Links:

- **Live Cloudflare Demonstration:** https://nickname-boundary-normal-map.trycloudflare.com
- **FastAPI Interactive Swagger Docs:** http://localhost:8000/docs
- **Node Express API Gateway:** http://localhost:3001/api/curriculum
- **SIH Portal:** https://www.sih.gov.in